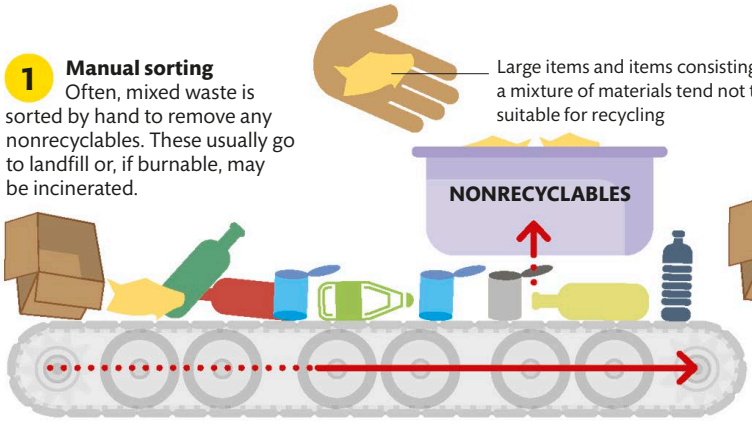
Recycling

Recycling is the process of collecting waste items and breaking them down into materials that can be turned into new products. A key part of the process is the sorting of items into their different materials, such as glass or plastic, so that they can be sent to the appropriate reprocessing facility.

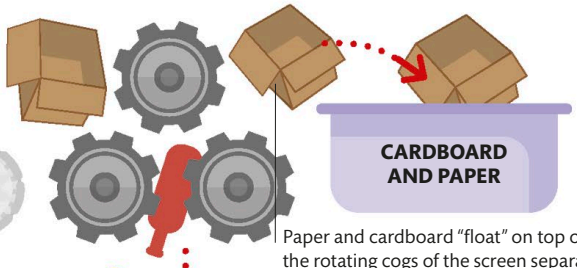
Materials' recovery

The sorting and cleaning of recyclables happens at a materials recovery facility (MRF). Through a combination of systems and processes that varies between MRFs, materials are recovered and forwarded to specialist plants for processing. Recyclable materials include paper and cardboard, which are made into new paper and card products, and glass, which is turned into new bottles and jars. Some items, such as electronics, that are complex and contain many different components are processed at specialist recycling facilities.



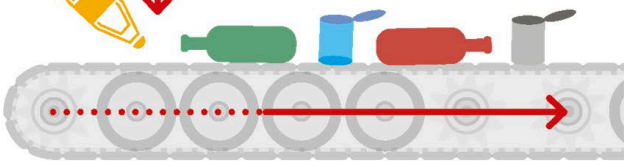
1 Manual sorting
Often, mixed waste is sorted by hand to remove any nonrecyclables. These usually go to landfill or, if burnable, may be incinerated.

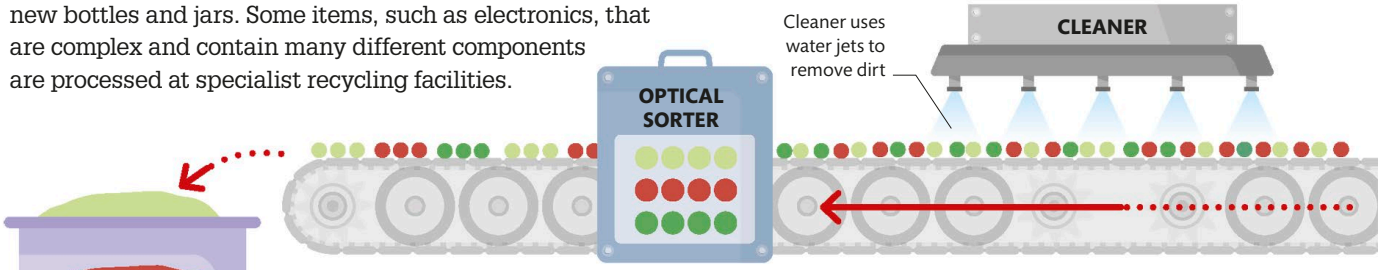
Large items and items consisting of a mixture of materials tend not to be suitable for recycling



2 Paper and cardboard recovery
Paper and cardboard are recovered by a screen system that separates them from heavier materials. They are sent to specialist plants for reprocessing into new items.

Paper and cardboard "float" on top of the rotating cogs of the screen separator; other materials fall straight through



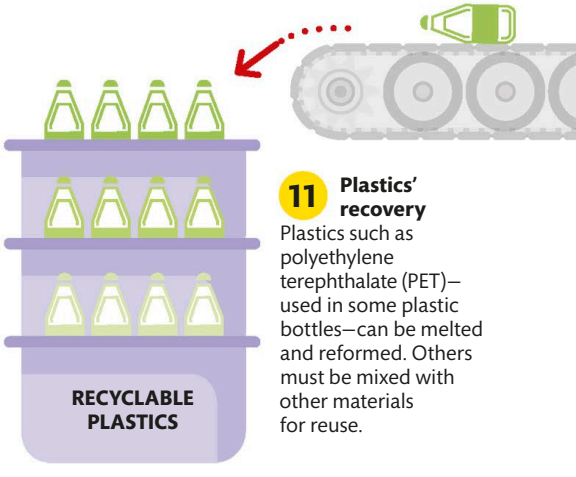


Cleaner uses water jets to remove dirt

7 Glass cleaned
The crushed glass is cleaned to remove any impurities. The cleaned glass may be sorted by color or used in products such as roadbed.

8 Glass sorted
Some glass recycling plants use advanced optical scanners to sort the glass fragments by color.

9 Glass recovery
Sorted glass may go on to be melted down and remolded into new bottles and jars or other glass items of uniform color.



11 Plastics' recovery
Plastics such as polyethylene terephthalate (PET)—used in some plastic bottles—can be melted and reformed. Others must be mixed with other materials for reuse.